[Big Data Analytics]

Pet Adoption Speed Prediction

DaeHan Ahn

AI Convergence

울산대학교
UNIVERSITY OF ULSAN

Adoption Speed Prediction Based on Animal Profile

- **Background**
  - Pets have become an important part of our lives and are often regarded as members of the family. However, many pets are still abandoned and left without homes. Around the world, animal welfare organizations rescue these animals and help them wait for new families.
  - In this project, we will analyze adoption data to identify what types of animals are more likely to be adopted quickly. We will also examine the effects of controllable factors, such as vaccination status, sterilization status, and the number of photos, to understand what kinds of adoption profiles may help improve the chances of adoption.

- **As-Is**
  - Adoption profiles are created using basic information such as age, species, color, and photos.
  - The adoption process begins when potential adopters show interest.
- **To-Be**
  - Analyze which animal characteristics are associated with faster adoption.
  - Separate features into **uncontrollable factors** and **controllable factors**.
  - Find how adoption profiles can be improved using controllable factors.
- **Goal**
  - Develop an **adoption speed prediction model** based on animal profiles.
  - Identify useful controllable factors, such as vaccination, sterilization, and number of photos.
  - Suggest profile strategies that may improve adoption chances.

Mission Workflow

- Follow the Markdown guidelines provided in the source code while completing the mission.
- Upload your source code to **Classum** after the class.
- The mission will be evaluated as **Pass/Fail** based on the leaderboard score.

- 소스코드 내 markdown 가이드라인을 따라 미션 수행 .
- 수업종료 후 소스코드 업로드 할것 (Classum).
- 미션 수행 Pass/Fail 여부는 리더보드 스코어를 보고 판단함.

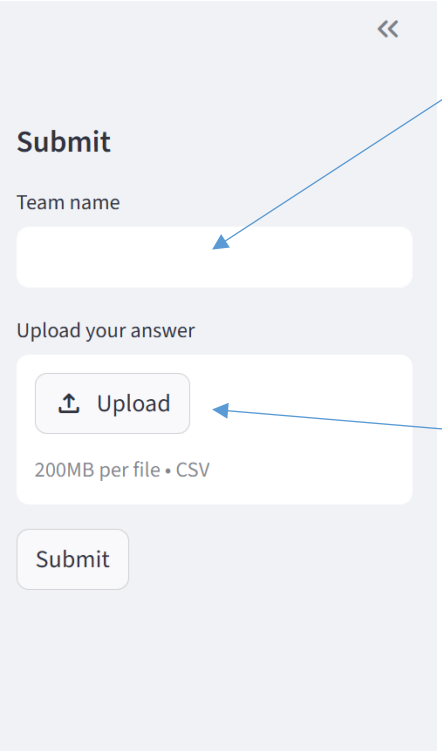
Rules

- **Restrictions**
  - The use of **Generative AI is strictly prohibited**
  - Searching **Data Science Competition Platforms** (e.g., Kaggle, Dacon, etc.) is prohibited
  - Any search related to analysis examples or solutions using the same dataset is prohibited
  - Any violation will result in a **score of zero**

- **Grading Policy**
  - **Pass/Fail** (60% weight)
  - **Presentation** (40% weight)
    - On Wednesday classes, the **top 3 teams must present on Thursday**, and additional points will be awarded based on presentation quality
      - Data analysis skills
      - Ability to derive meaningful insights
    - Presentations may be conducted using a **Jupyter Notebook**
    - Other teams will also have opportunities to present within the remaining time, and additional points will be awarded
    - Additional analysis can be conducted between Wednesday and Thursday

File Upload

- **Link:** https://bigdataanalysis-dashboard.streamlit.app/
- **File Format:** CSV file only
- **Contents:** the model predictions for test.csv



Team name: it may remain anonymous; however, the name must be revealed in the final round. Scores will not be recorded if the team number is not disclosed.

	A
1	prediction
2	2
3	1
4	2
5	2

Column name must be ‘prediction’

Thank you !

Q & A